E-Commerce Application Testing

DEPI - Digital Egypt Pioneers Initiative | Track: Software Testing

1. Project Planning & Management

Project Proposal

This project encompasses comprehensive manual and automated testing for a new e-commerce application. The primary objective is to guarantee functional integrity, robust performance, and optimal user satisfaction prior to market launch.

Project Timeline & Execution Strategy

The engagement follows an 8-week timeline designed to cover all key software testing lifecycle stages:

- **Requirement Analysis:** Review and clarify functional and non-functional specifications.
- **Test Planning:** Define test scope, strategy, resources, and schedules.
- **Test Case Development:** Design manual scripts and automated test suites.
- **Environment Setup:** Configure staging servers, test databases, and automation frameworks.
- **Test Execution:** Conduct functional, API, automated UI, and regression cycles.
- **Test Closure:** Final metrics evaluation, bug sign-off, and summary reporting.

Team Roles & Task Assignment

Team Member	Assigned Role	Primary Responsibilities
Abdullah Yasser	QA Lead / Test Manager	Overseeing ISTQB compliance, strategic

Team Member	Assigned Role	Primary Responsibilities
		planning, and overall test execution.
Senior Automation Engineer	Automation Specialist	Focus on Selenium UI scripting and framework maintenance.
Automation Engineer	API & Backend Specialist	Focus on backend and API validation via Postman.
Manual Test Engineer 1	Manual QA Specialist	Exploratory, functional, and user interface testing.
Manual Test Engineer 2	Compatibility Specialist	Cross-browser, localization, and mobile compatibility testing.
Defect Manager / QA Analyst	Jira Administrator	Defect triage, metric tracking, and process flow management.

Risk Assessment & Mitigation Plan

Risk Description	Severity	Mitigation Strategy
UI Instability: Frequent UI changes breaking Selenium scripts.	High	Implement a robust Page Object Model (POM) design pattern to isolate UI changes.
Environment Downtime: Unstable test environment	Medium	Schedule dedicated maintenance windows

Risk Description	Severity	Mitigation Strategy
impacting execution schedule.		outside of peak execution hours.

Key Performance Indicators (KPIs)

- **Defect Density:** Total defects identified relative to the size of the module.
- **Test Execution Coverage:** Percentage of planned test cases executed against requirements.
- **Automation Script Pass Rate:** Ratio of successful automated runs during regression suites.
- **Defect Leakage:** Percentage of invalid or missed defects identified post-cycle.

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2. Literature Review

Feedback & Evaluation

Initial reviews confirm a strong baseline test strategy aligned with standard ISTQB guidelines. Synthesizing Postman for backend API validation alongside Selenium for frontend automation provides balanced, multi-layered test coverage across the application stack.

Suggested Improvements

To further optimize the quality engineering workflow, it is recommended to enhance CI/CD pipeline integration. Continuous integration will enable daily, unattended automated test executions, allowing the team to identify and resolve regressions significantly earlier in each sprint cycle.

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3. Requirements Gathering

Stakeholder Analysis

- **Product Owners:** Concerned with business logic alignment, flow correctness, and operational readiness.
- **Developers:** Expect detailed, actionable, and easily reproducible bug reports within Jira.
- **End-Users:** Require seamless usability, fast response times, and uncompromising security.

User Stories & Core Use Cases

User Story 1

"As a customer, I want to add items to my cart so that I can purchase them later."

- **Primary Use Case:** Add to Cart functionality, quantity adjustments, and persistent cart state validation.

User Story 2

"As a registered user, I want to securely check out using multiple payment methods."

- **Primary Use Case:** End-to-end Secure Checkout Process, transaction processing, and order confirmation generation.

System Requirements Summary

Requirement Category	Scope & Specifications
Functional Requirements	User Authentication, Product Catalog Search, Cart Management, Payment Gateway Integration, Order Tracking.
Non-Functional Requirements	**Performance:** Page load speed under 3 seconds.**Security:** End-to-end data

Requirement Category	Scope & Specifications
	encryption in transit and at rest.**Usability:** Fully responsive design across target mobile and desktop viewports.**Reliability:** High availability with 99.9% target uptime.